CHALLENGE

2 Points (Individual · Optional)

State Machine

LEARNING OBJECTIVES

You design and implement a **finite state machine (FSM)** for a game scenario. You learn to structure complex behaviour through clearly defined states and transitions.

01 Design

Design a finite state machine for a game scenario with at least three states. Document the states and their transition conditions in a diagram or table.

One possible scenario is a weather system with states such as **Sunny**, **Cloudy**, **Rainy** and **Stormy**. Transitions can be triggered by variables like temperature or wind speed, or by timers. Each state should be clearly distinguishable visually or through output.

02 Implementation

Implement the designed state machine so that the object transitions between states automatically and each state shows clearly distinguishable behaviour.

This assignment can be implemented in a game engine or any other programming environment. A simple visual representation or console output is sufficient.

RESOURCES

Game Programming Patterns: State

ACCEPTANCE REQUIREMENTS

- [ ] States and transition conditions are documented as a diagram or table.
- [ ] The state machine has at least three states; the object transitions between them automatically and each state shows clearly distinguishable behaviour.